



DW02_mac

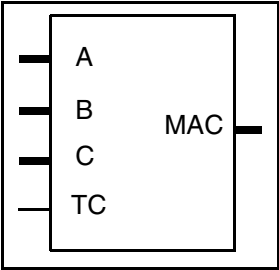
Multiplier-Accumulator

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Features and Benefits

- Parameterized word length
- Unsigned and signed (two's-complement) data operation
- Inferable using a function call

Revision History



Description

DW02_mac is a multiplier-accumulator. It multiplies a number *A* by a number *B*, and adds the result to a number *C* to produce a result *MAC*.

The input control signal *TC* determines whether the inputs and outputs are interpreted as unsigned (*TC* = 0) or signed (*TC* = 1) numbers.

To extend the accuracy of the accumulator beyond $A \times B$, use DW02_prod_sum1 Multiplier-Adder in place of DW02_mac.

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
A	<i>A_width</i> bits	Input	Multiplier
B	<i>B_width</i> bits	Input	Multiplicand
C	<i>A_width</i> + <i>B_width</i> bits	Input	Addend
TC	1 bit	Input	Two's complement control ■ 0 = Unsigned ■ 1 = Signed
MAC	<i>A_width</i> + <i>B_width</i> bits	Output	MAC result ($A \times B + C$)

Table 1-2 Parameter Description

Parameter	Values	Description
A_width	≥ 1	Word length of <i>A</i>
B_width	≥ 1	Word length of <i>B</i>

Table 1-3 Synthesis Implementations^a

Implementation Name	Function	License Feature Required
pparch	Delay-optimized flexible Booth Wallace	DesignWare
apparch	Area-optimized flexible architecture that can be optimized for area, for speed, or for area, speed	DesignWare

a. During synthesis, Synopsys synthesis tools select the appropriate architecture for your constraints. Alternatively, you may use the `set_implementation` command to choose one specific implementation from those listed in this table. For details, see the documentation for your Synopsys synthesis tool.

Table 1-4 Simulation Models

Model	Function
DW02.DW02_MAC_CFG_SIM	Design unit name for VHDL simulation
dw/dw02/src/DW02_mac_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW02_mac.v	Verilog simulation model source code

Table 1-5 Functional Description

TC	A	B	MAC
0	A (unsigned)	B (unsigned)	$(A \times B) + C$ (unsigned)
1	A (two's complement)	B two's complement	$(A \times B) + C$ (two's complement)

Related Topics

- [Math – Arithmetic Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Function Inferencing - VHDL

```
library IEEE, DWARE;
use IEEE.std_logic_1164.all;
use IEEE.std_logic_arith.all;
use DWARE.DW_foundation_arith.all;

entity DW02_mac_func is
  generic(wordlength1 : integer:=8; wordlength2: integer := 8);
  port(func_A    : in std_logic_vector(wordlength1-1 downto 0);
       func_B    : in std_logic_vector(wordlength2-1 downto 0);
       func_C    : in std_logic_vector(wordlength1+wordlength2-1 downto 0);
       func_TC   : in std_logic;
       MAC_func  : out std_logic_vector(wordlength1+wordlength2-1 downto 0) );
end DW02_mac_func;

architecture func of DW02_mac_func is
begin

  process (func_A,func_B,func_C, func_TC)
  begin
    if func_TC = '1' then
      MAC_func <= std_logic_vector(DWF_mac(SIGNED(func_A),
                                           SIGNED(func_B), SIGNED(func_C)) );
    else
      MAC_func <= std_logic_vector(DWF_mac(UNSIGNED(func_A),
                                           UNSIGNED(func_B), UNSIGNED(func_C)) );
    end if;
  end process;
end func;
```

HDL Usage Through Function Inferencing - Verilog

```
module DW02_mac_func (func_A, func_B, func_C, func_TC, MAC_func);
    parameter func_A_width = 8;
    parameter func_B_width = 8;

    // Passes the widths to the multiplier-accumulator function
    parameter A_width = func_A_width;
    parameter B_width = func_B_width;

    // Please add search_path = search_path + {synopsys_root + "/dw/sim_ver"}
    // to your .synopsys_dc.setup file (for synthesis) and add
    // +incdir+$SYNOPSYS/dw/sim_ver+ to your verilog simulator command line
    // (for simulation).
    `include "DW02_mac_function.inc"

    input [func_A_width-1 : 0]          func_A;
    input [func_B_width-1 : 0]          func_B;
    input [func_A_width+func_B_width-1 : 0] func_C;
    input                                func_TC;

    output [func_A_width+func_B_width-1 : 0] MAC_func;

    assign MAC_func = (func_TC) ? DWF_mac_tc(func_A, func_B, func_C) :
                             DWF_mac_uns(func_A, func_B, func_C);

endmodule
```

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_foundation_comp.all;

entity DW02_mac_inst is
  generic ( inst_A_width : NATURAL := 8;
            inst_B_width : NATURAL := 8 );
  port (inst_A  : in std_logic_vector(inst_A_width-1 downto 0);
        inst_B  : in std_logic_vector(inst_B_width-1 downto 0);
        inst_C  : in std_logic_vector(inst_A_width+inst_B_width-1 downto 0);
        inst_TC : in std_logic;
        MAC_inst: out std_logic_vector(inst_A_width+inst_B_width-1 downto 0) );
end DW02_mac_inst;

architecture inst of DW02_mac_inst is
begin

  -- Instance of DW02_mac
  U1 : DW02_mac
    generic map ( A_width => inst_A_width, B_width => inst_B_width )
    port map ( A => inst_A, B => inst_B, C => inst_C,
              TC => inst_TC, MAC => MAC_inst );
end inst;

-- pragma translate_off
configuration DW02_mac_inst_cfg_inst of DW02_mac_inst is
  for inst
  end for; -- inst
end DW02_mac_inst_cfg_inst;
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

```
module DW02_mac_inst( inst_A, inst_B, inst_C, inst_TC, MAC_inst );

    parameter A_width = 8;
    parameter B_width = 8;

    input [A_width-1 : 0] inst_A;
    input [B_width-1 : 0] inst_B;
    input [A_width+B_width-1 : 0] inst_C;
    input inst_TC;
    output [A_width+B_width-1 : 0] MAC_inst;

    // Instance of DW02_mac
    DW02_mac #(A_width, B_width)
        U1 ( .A(inst_A), .B(inst_B), .C(inst_C), .TC(inst_TC), .MAC(MAC_inst) );

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
March 2019	DWBB_201903.0	■ Removed Table 1-4, “Obsolete Synthesis Implementations”
January 2019	DWBB_201806.5	■ Updated example in “ HDL Usage Through Component Instantiation - VHDL ” on page 5 ■ Added this Revision History table and the document links on this page

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